

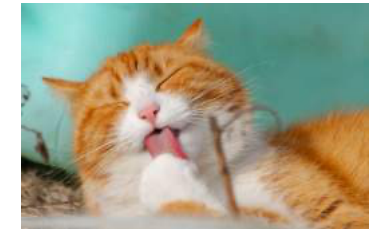


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Introduction to ML strategy

Why ML Strategy?

Motivating example



90%

Ideas:

- Collect more data ←
- Collect more diverse training set
- Train algorithm longer with gradient descent
- Try Adam instead of gradient descent
- Try bigger network
- Try smaller network
- Try dropout
- Add L_2 regularization
- Network architecture
 - Activation functions
 - # hidden units
 - ...

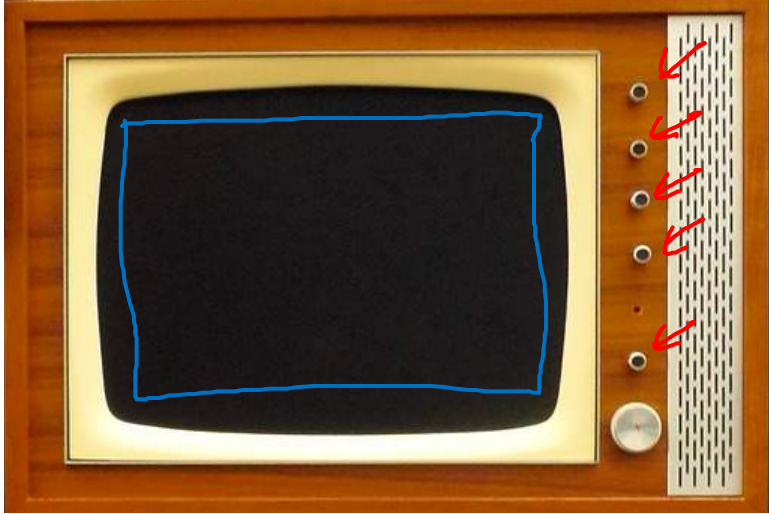


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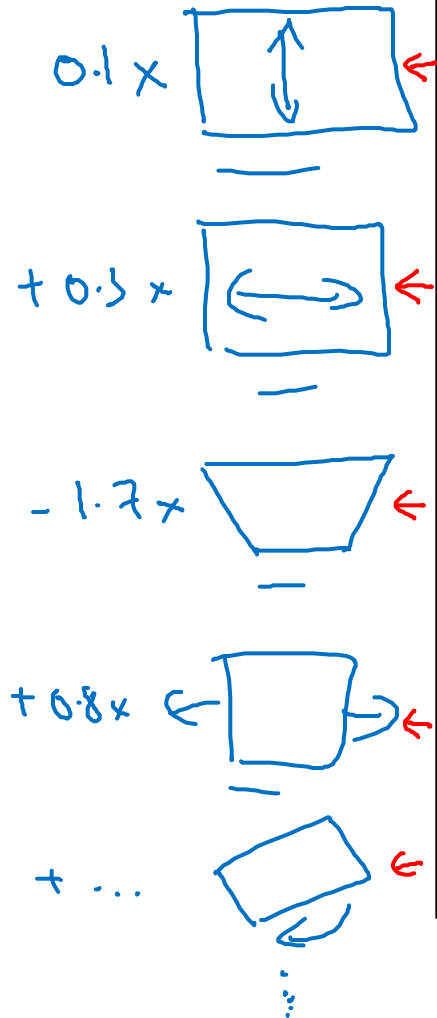
Introduction to ML strategy

Orthogonalization

TV tuning example



Orthogonalization



Car

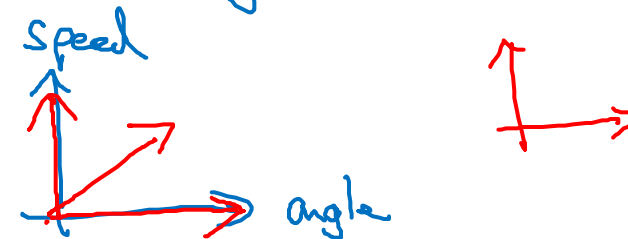


$\rightarrow$ Steering

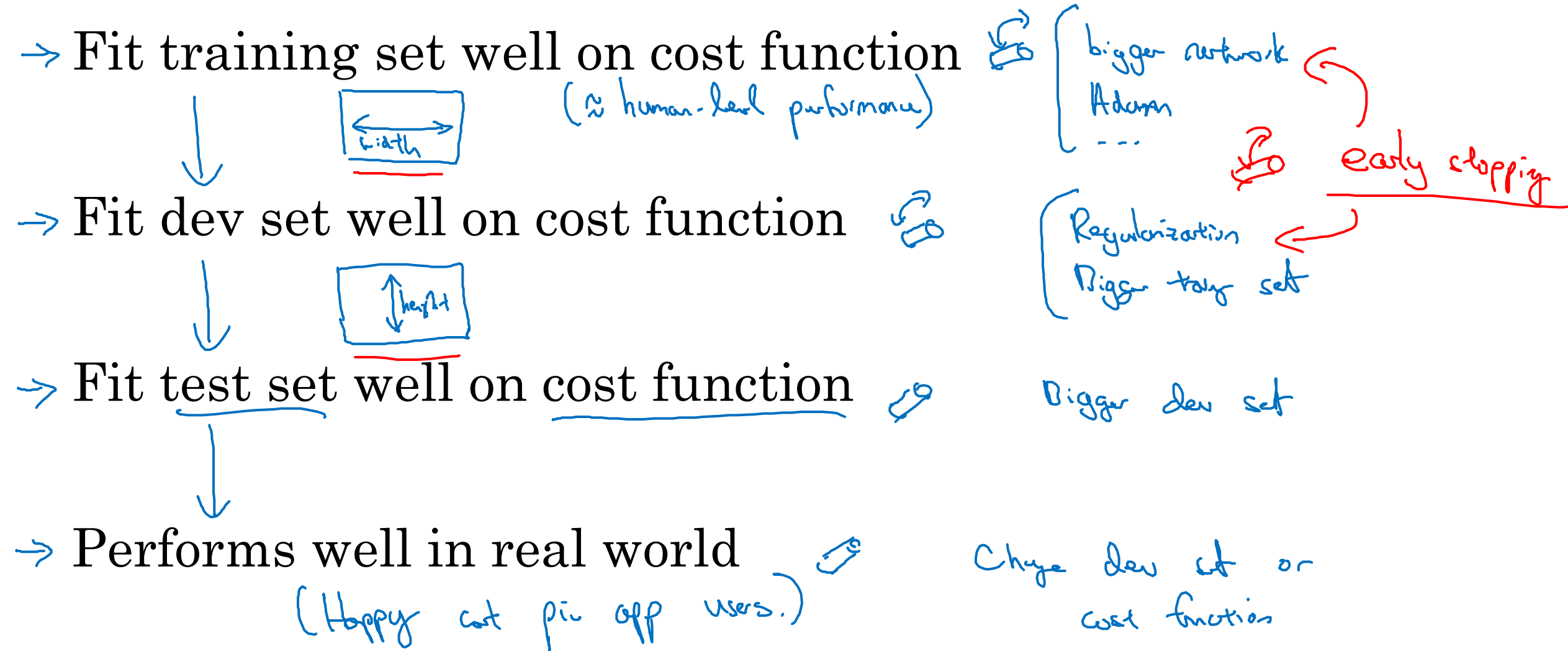
$\rightarrow$ { Accelerator
Braking }

$$\rightarrow \frac{0.3 \times \text{angle} - 0.8 \text{ speed}}{}$$

$$\rightarrow 2 \times \text{angle} + 0.9 \text{ speed}$$



Chain of assumptions in ML



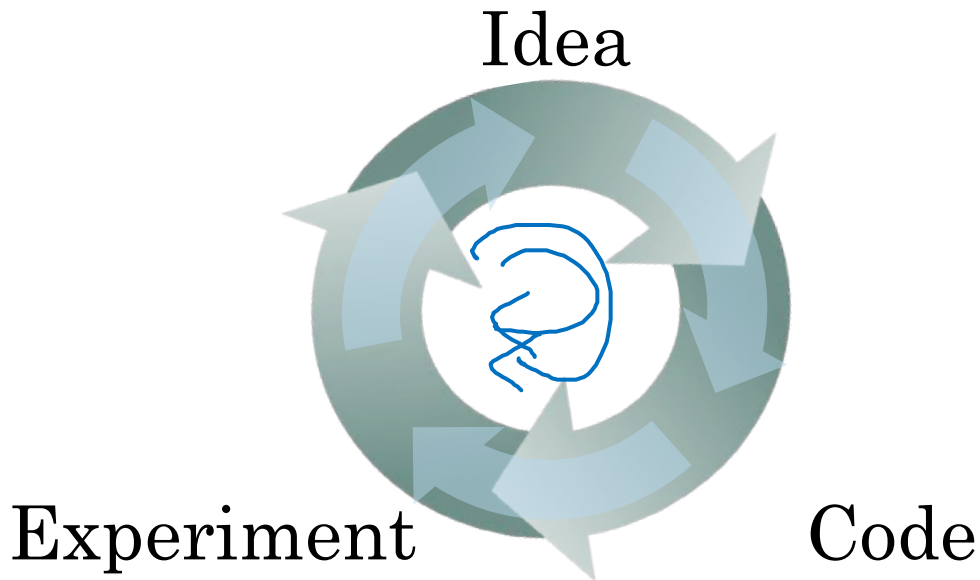


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Setting up
your goal

Single number
evaluation metric

Using a single number evaluation metric



→ Of examples recognized as cost, what % actually are costs?

→ what % of actual costs are correctly recognized

Classifier	Precision	Recall
A	95%	90%
B	98%	85%

F₁ score = "Average" of P and R.

$$\left(\frac{2}{\frac{1}{P} + \frac{1}{R}} \right) \text{ "Harmonic mean"}$$

Dev set + Single number evaluation metric
real speed up iterating

Another example

Algorithm	US	China	India	Other
A	<u>3%</u>	7%	5%	9%
B	5%	6%	5%	10%
C	2%	3%	4%	5%
D	5%	8%	7%	2%
E	4%	5%	2%	4%
F	7%	11%	8%	12%





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Setting up
your goal

Satisficing and
optimizing metrics

Another cat classification example

Classifier	Accuracy	Running time
A	90%	80ms
B	92%	95ms
C	95%	1,500ms

$$\text{Cost} = \text{accuracy} - 0.5 \times \text{Running Time}$$

maximize accuracy

subject to Running Time $\leq$ 100 ms.

N metrics : 1 optimizing
N-1 satisficing

Wakewords / Trigger words

Alexa, OK Google,

Hey Siri, nihao baidu
你好 百度

accuracy.

#false positive

maximize accuracy.

s.t. $\leq$ 1 false positive
every 24 hours.



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Setting up
your goal

Train/dev/test
distributions

Cat classification dev/test sets

development set, hold out cross validation set

Regions:

- US
- UK
- Other Europe
- South America
- India
- China
- Other Asia
- Australia

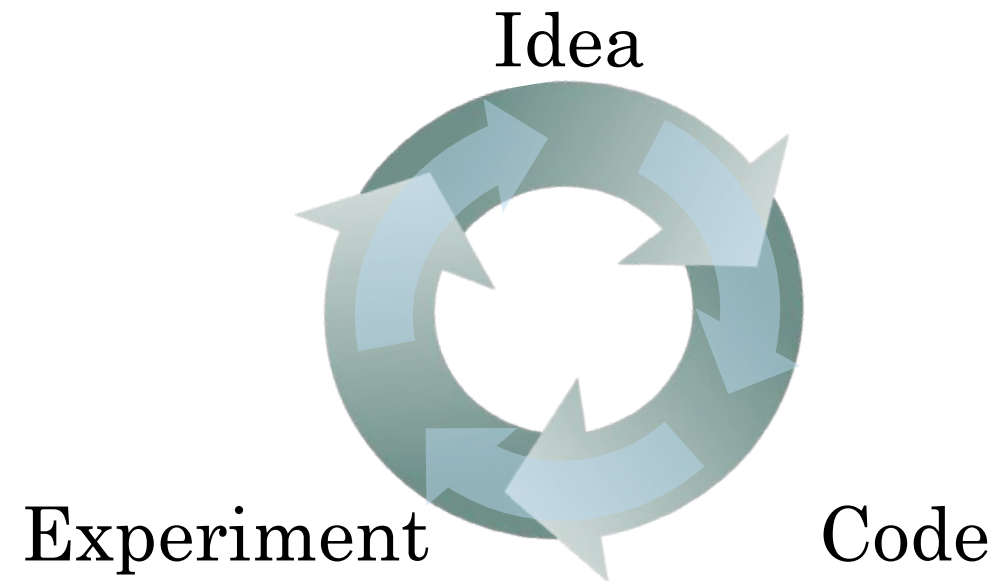
Dev

Test

→ Randomly shuffle into dev/test



dev set
+
metric



True story (details changed)

[Optimizing on dev set on loan approvals for
medium income zip codes

↑

$x \rightarrow y$ (repay loan?)



[Tested on low income zip codes

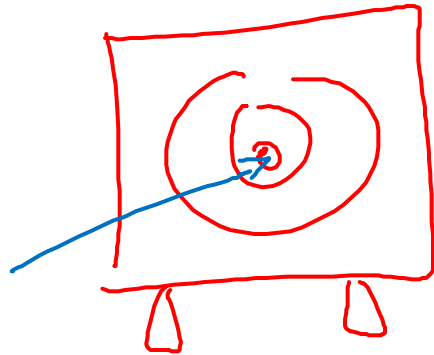
~ 3 month



Guideline

Choose a dev set and test set to reflect data you expect to get in the future and consider important to do well on.

training



dev
metric

test

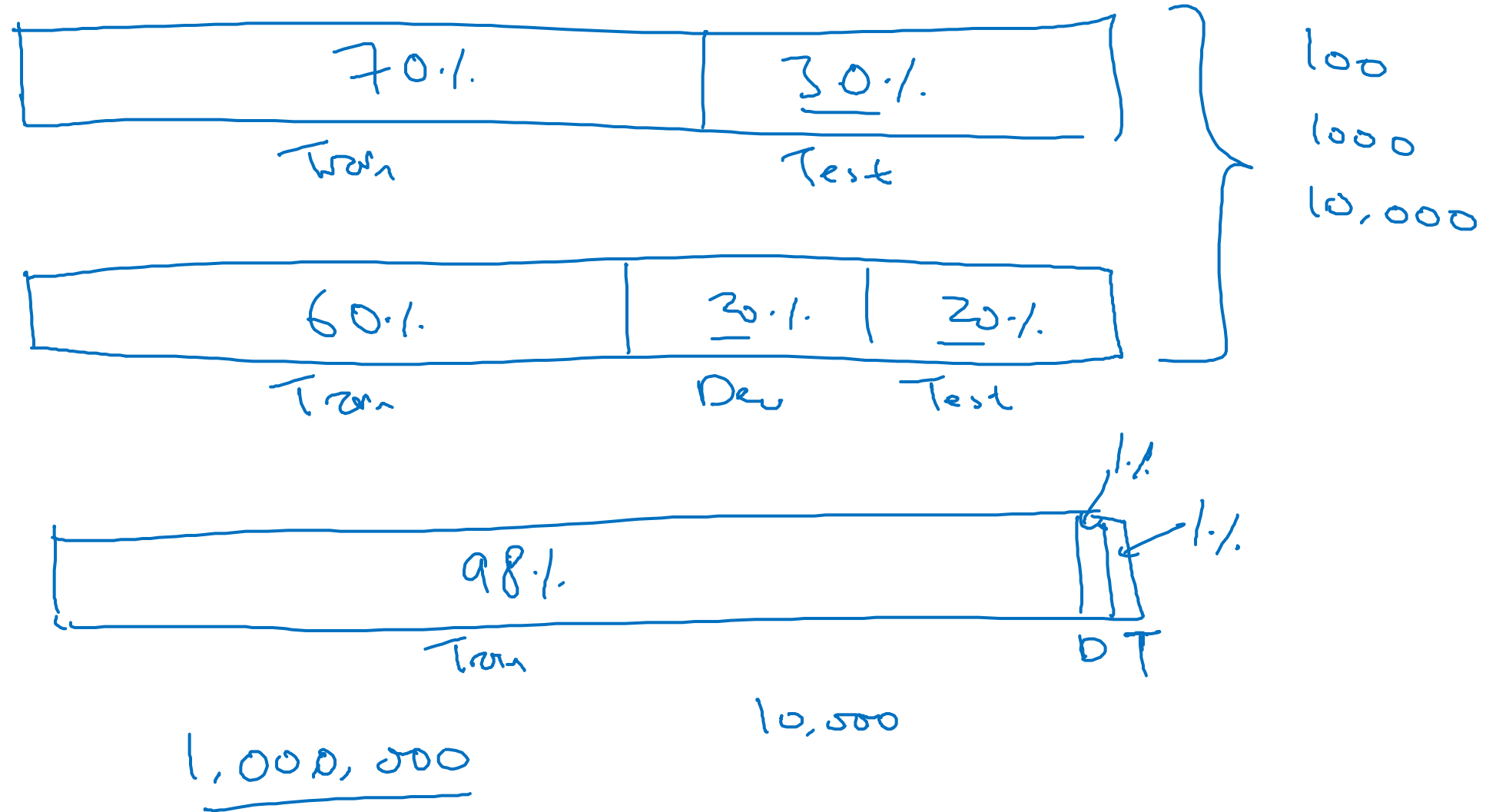


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Setting up
your goal

Size of dev
and test sets

Old way of splitting data



Size of dev set

A B

Set your dev set to be big enough to detect differences in
algorithm/models you're trying out.

100 : small
└ 1%

1,000

10,000

100,000

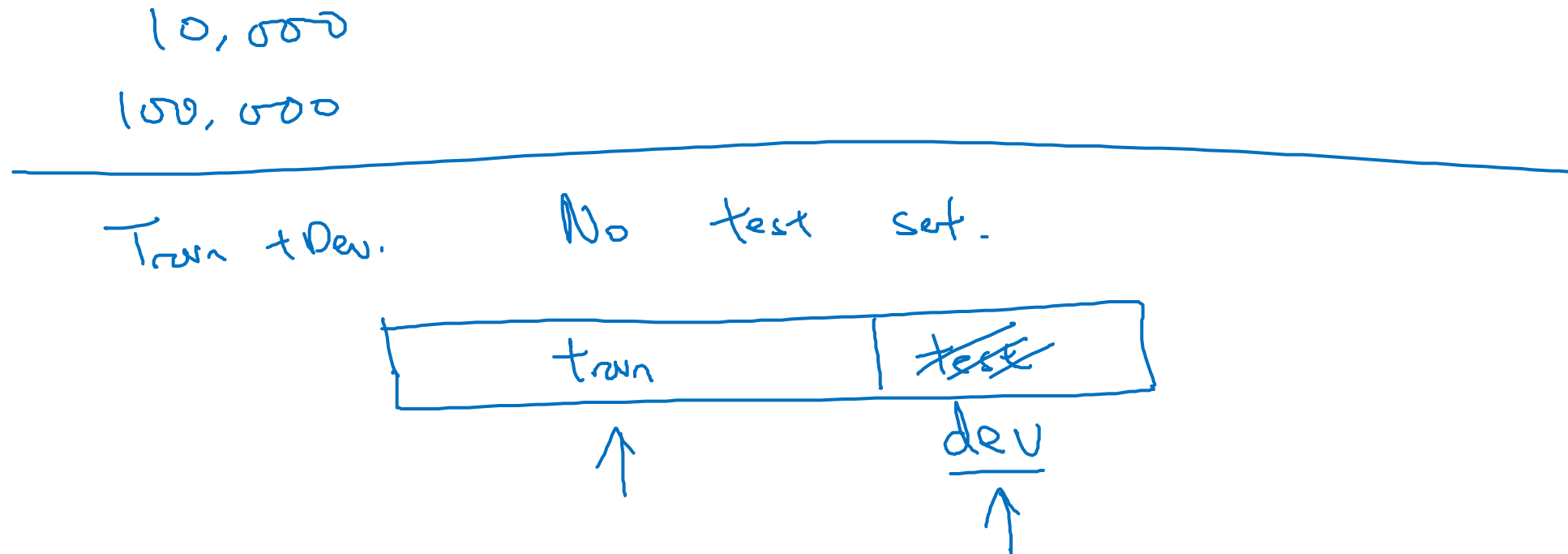
^A 97% → ^B 97.1%
0.1%
└

0.01%
└
0.001%

Online advertising

Size of test set

- Set your test set to be big enough to give high confidence in the overall performance of your system.





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Setting up
your goal

When to change
dev/test sets and
metrics

Cat dataset examples

Metric + Dev : Prefer A
You/users : Prefer B.

→ Metric: classification error

Algorithm A: 3% error

→ pornographic

✓ Algorithm B: 5% error

Error: $\frac{1}{\sum_i w^{(i)}} \cdot \frac{1}{m_{dev}} \sum_{i=1}^{m_{dev}} w^{(i)} \mathbb{I}\{y_{pred}^{(i)} \neq y^{(i)}\}$

↪ $w^{(i)} = \begin{cases} 1 & \text{if } x^{(i)} \text{ is non-porn} \\ 10 & \text{if } x^{(i)} \text{ is porn} \end{cases}$

$\mathbb{I}\{y_{pred}^{(i)} \neq y^{(i)}\}$
predicted value (0/1)

Orthogonalization for cat pictures: anti-porn

- 1. So far we've only discussed how to define a metric to evaluate classifiers. ← Place target ↗
- 2. Worry separately about how to do well on this metric. ↗
- ↖ Aim (shoot at target)

$$\rightarrow J = \frac{1}{\sum w^{(i)}} \sum_{i=1}^m w^{(i)} \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$



Another example

Algorithm A: 3% error

✓ Algorithm B: 5% error ←

→ Dev/test



→ User images



If doing well on your metric + dev/test set does not correspond to doing well on your application, change your metric and/or dev/test set.

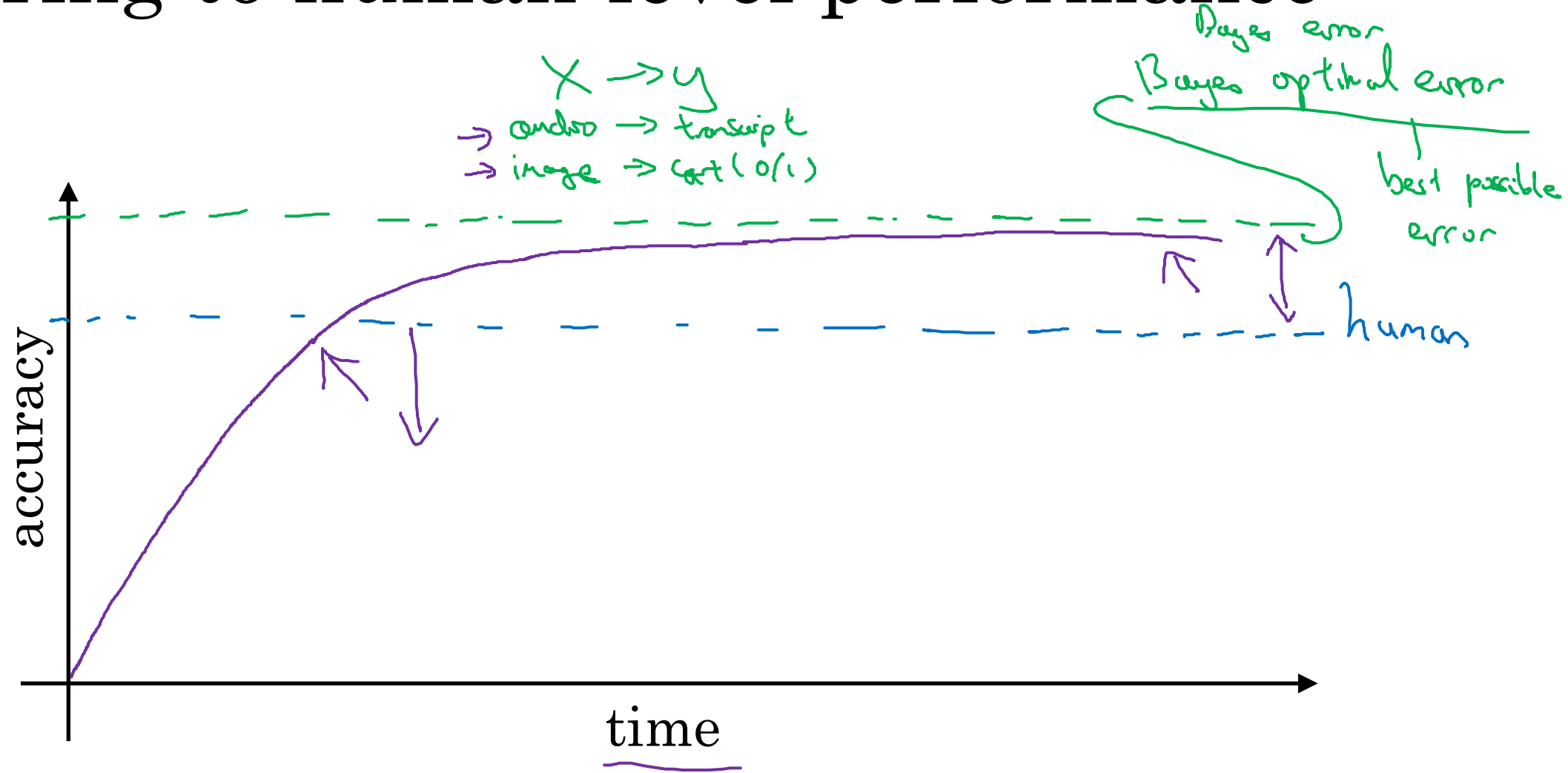


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Comparing to human-
level performance

Why human-level
performance?

Comparing to human-level performance



Why compare to human-level performance

Humans are quite good at a lot of tasks. So long as ML is worse than humans, you can:

- - Get labeled data from humans. (x, y)
- - Gain insight from manual error analysis:
Why did a person get this right?
- - Better analysis of bias/variance.

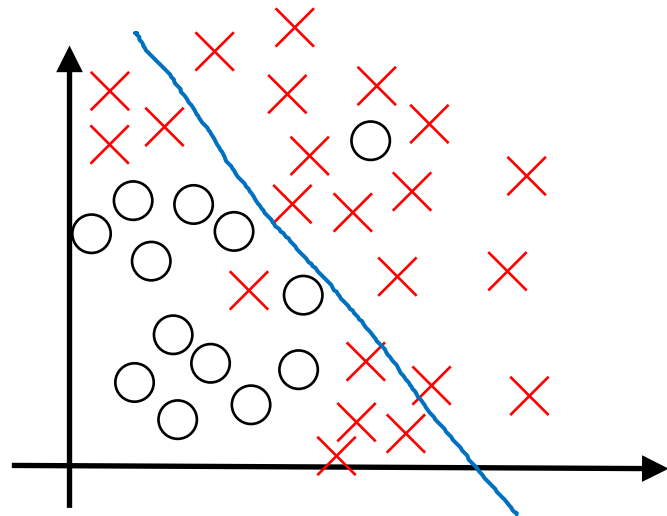


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Comparing to human-
level performance

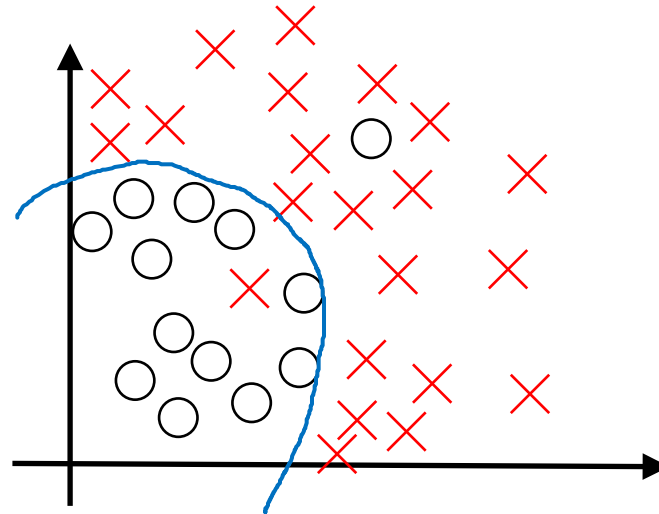
Avoidable bias

Bias and Variance

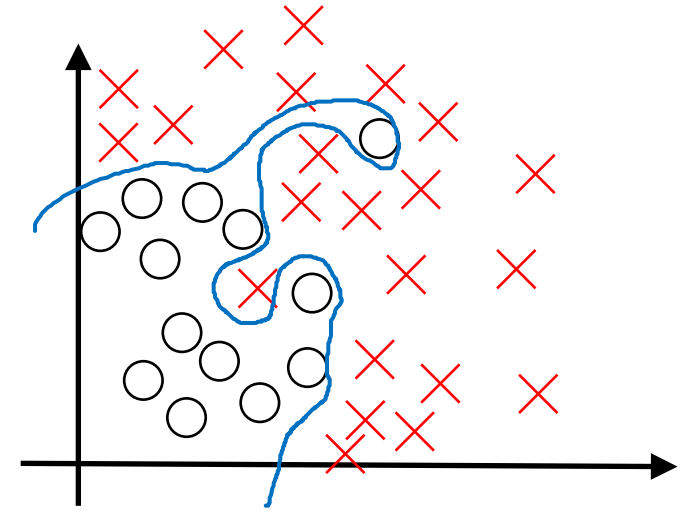


high bias

underfitting



“just right”



high variance

overfitting

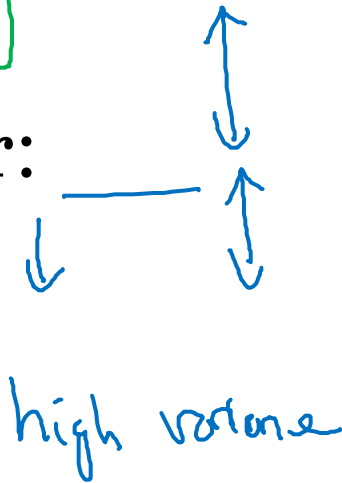
Bias and Variance

Cat classification

Human-level $\approx 0\%$ ----

Training set error:

Dev set error:



high variance

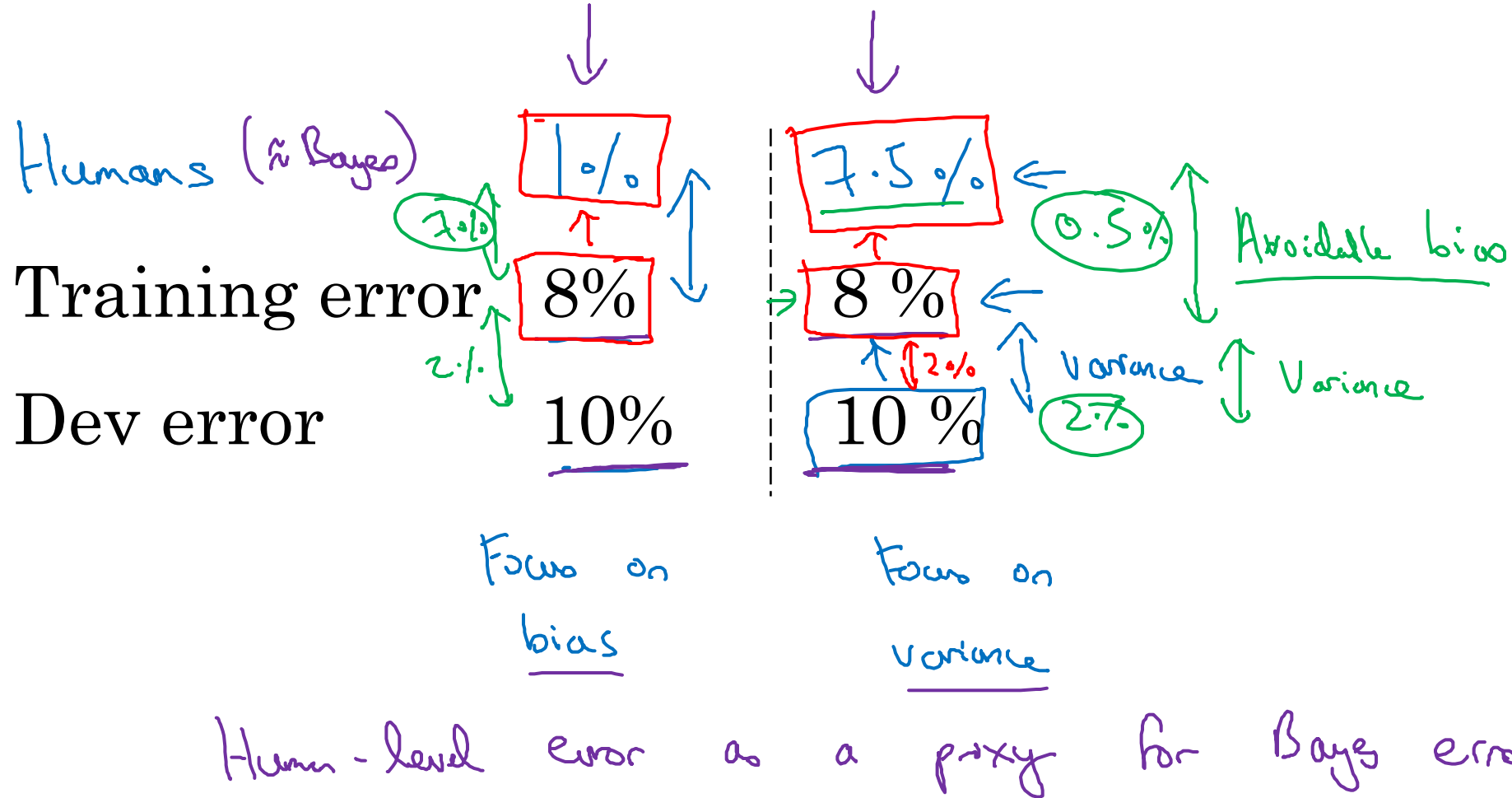


high bias

high bias
high variance

low bias
low variance

Cat classification example





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Comparing to human-level performance

Understanding
human-level
performance

Human-level error as a proxy for Bayes error

Medical image classification example:

Suppose:

(a) Typical human 3 % error

→ (b) Typical doctor 1 % error

(c) Experienced doctor 0.7 % error

→ (d) Team of experienced doctors .. 0.5 % error ←

Bayes error $\leq$ 0.5 %

What is “human-level” error?



Error analysis example

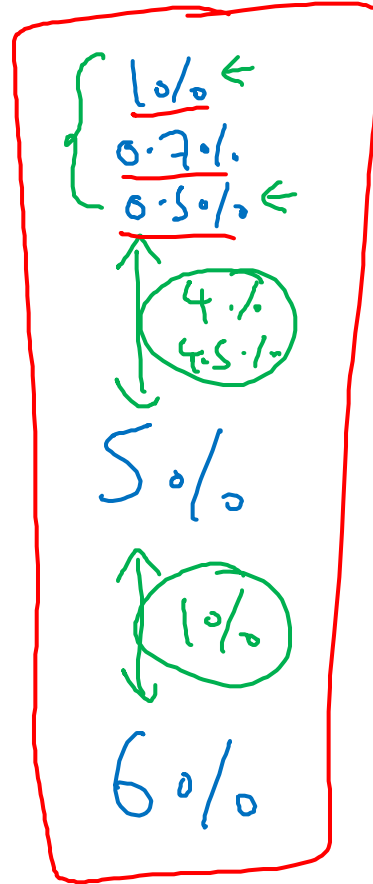
Human (proxy for Bayes error)

↑ Avoidable bias

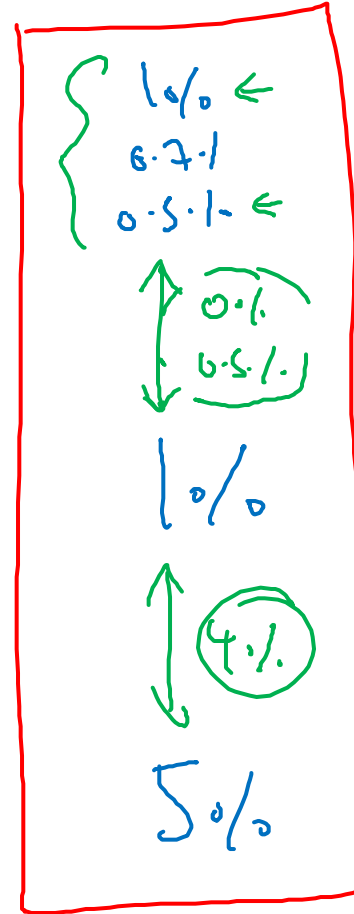
Training error

↑ Variance

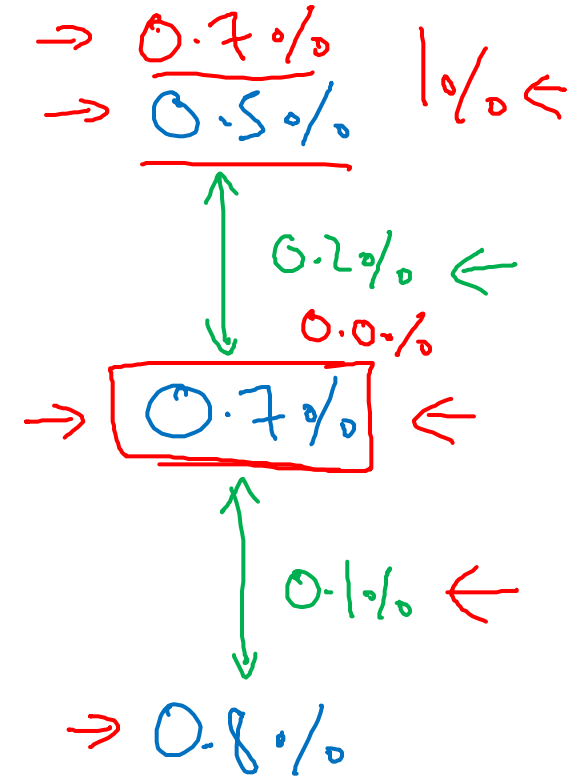
Dev error



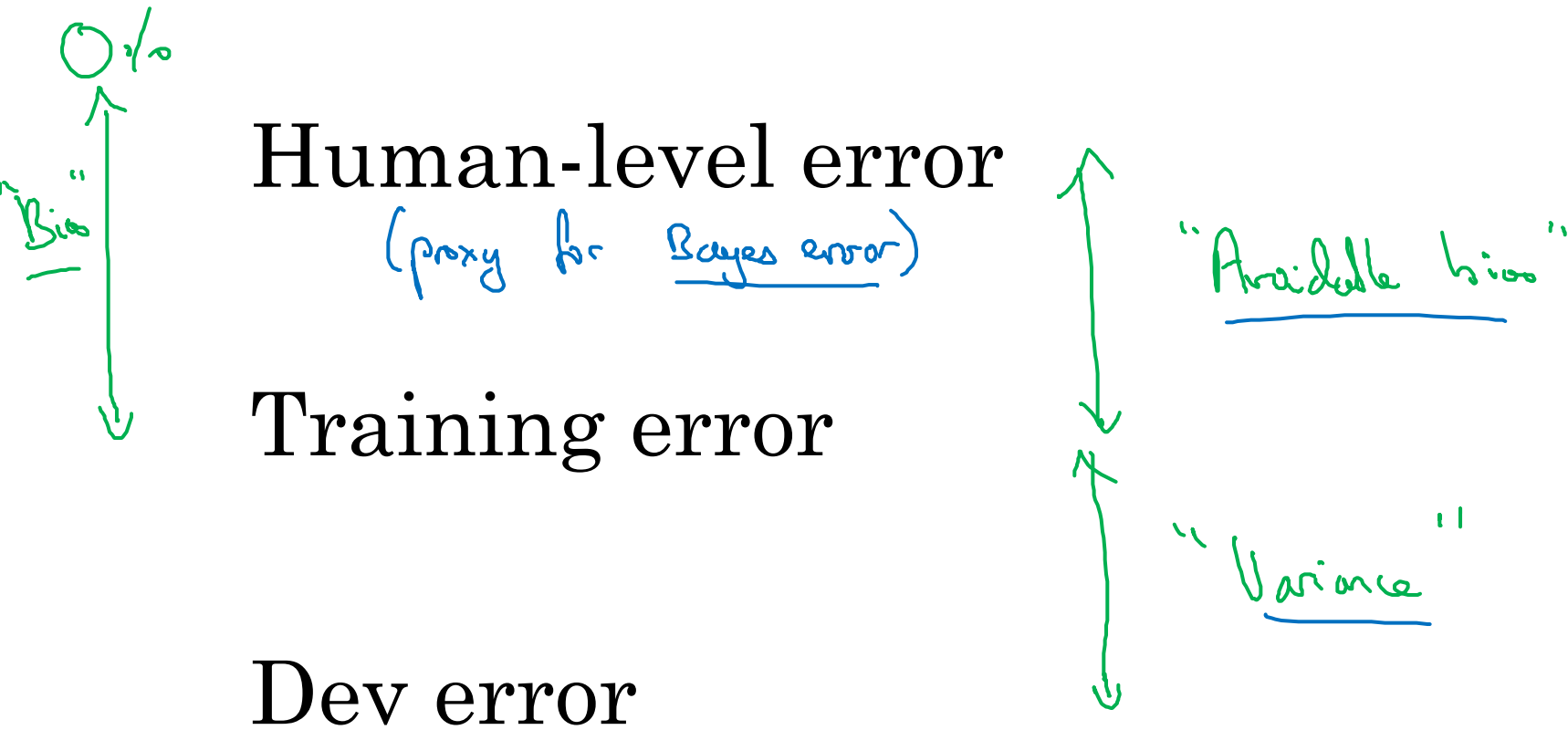
↑ Bias



↑ Variance



Summary of bias/variance with human-level performance





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Comparing to human-
level performance

Surpassing human-
level performance

Surpassing human-level performance

Team of humans

0.5%

One human

0.1 ~~1.0%~~

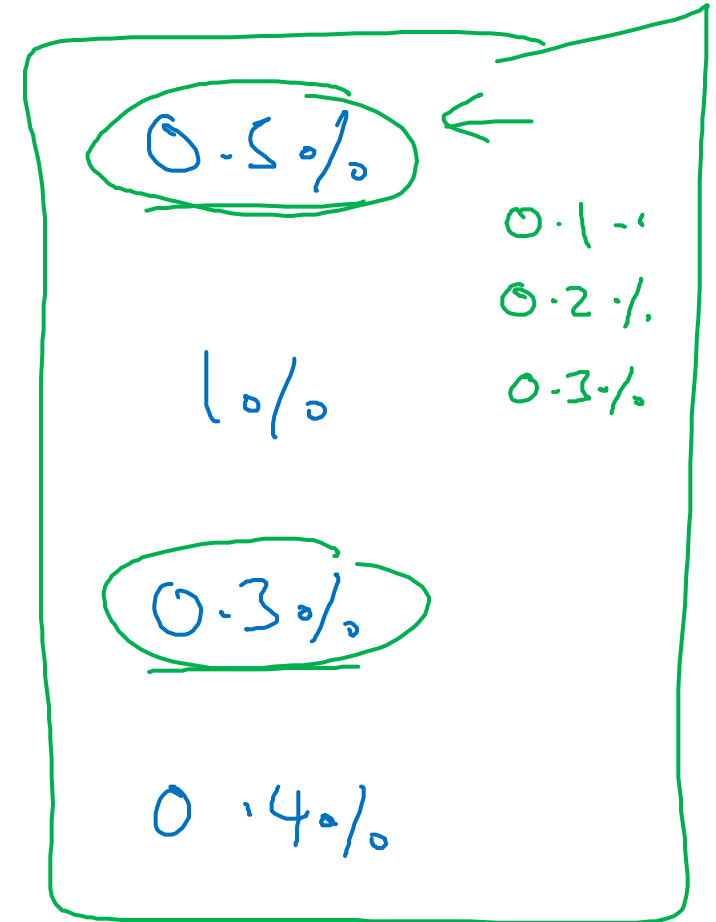
Training error

0.6%

Dev error

0.2
0.8%

What is avoidable bias?



Problems where ML significantly surpasses human-level performance

- - Online advertising
- - Product recommendations
- - Logistics (predicting transit time)
- - Loan approvals

Structured data

Not natural perception

Lots of data

- Speech recognition
- Some image recognition
- Medical
 - ECG, Skin cancer, ...



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Comparing to human-
level performance

Improving your model
performance

The two fundamental assumptions of supervised learning

1. You can fit the training set pretty well.



~ Avoidable bias

2. The training set performance generalizes pretty well to the dev/test set.



~ Variance

Reducing (avoidable) bias and variance

Human-level



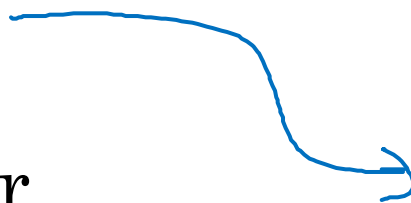
Avoidable bias



Training error



Variance



Dev error

Train bigger model

Train longer/better optimization algorithms

- momentum, RMSprop, Adam

NN architecture/hyperparameters search

RNN
CNN

More data

Regularization

- L_2 , dropout, data augmentation

NN architecture/hyperparameters search



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Error Analysis

Carrying out error
analysis

Look at dev examples to evaluate ideas



90% accuracy
→ 10% error

Should you try to make your cat classifier do better on dogs? ←

Error analysis:

- Get ~100 mislabeled dev set examples. → 5-10 min
- Count up how many are dogs.

→ 5%
5/100

10%
↓
9.5%

"ceiling"

→ 50%
50/100

10%
↓
5%

Evaluate multiple ideas in parallel

Ideas for cat detection:

- Fix pictures of dogs being recognized as cats ←
- Fix great cats (lions, panthers, etc..) being misrecognized ←
- Improve performance on blurry images ←

Image	Dog	Great Cats	Blurry	Instagram	Comments
1	✓			✓	Pitbull
2			✓	✓	
3		✓	✓		Rainy day at zoo
⋮	⋮	⋮	⋮	⋮	
% of total	<u>8%</u>	<u>43%</u>	<u>61%</u>	<u>12%</u>	



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Error Analysis

Cleaning up Incorrectly labeled data

Incorrectly labeled examples

x							
y	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	1

Training Set.

↑

DL algorithms are quite robust to random errors in the training set.

Systematic errors

Error analysis

✓

Image	Dog	Great Cat	Blurry	Incorrectly labeled	Comments
...					
98				✓	Labeler missed cat in background
99		✓			
100				✓	Drawing of a cat; Not a real cat.
% of total	<u>8%</u>	<u>43%</u>	<u>61%</u>	<u>6%</u>	

↑
↓

←

←

Overall dev set error 10%

Errors due incorrect labels 0.6% ←

Errors due to other causes 9.4% ←

↑

2.1%

1.9%

2.0%
0.6%
1.4%

Goal of dev set is to help you select between two classifiers A & B.

Correcting incorrect dev/test set examples

- Apply same process to your dev and test sets to make sure they continue to come from the same distribution
- Consider examining examples your algorithm got right as well as ones it got wrong. 20%
- Train and dev/test data may now come from slightly different distributions.

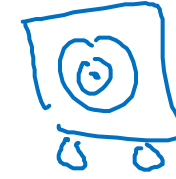


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Error Analysis

Build your first system
quickly, then iterate

Speech recognition example



- • Noisy background
 - • Café noise
 - • Car noise

- • Accent
- • Far from
- • Young
- • Stutter
- • ...

Guideline:

**Build your first
system quickly,
then iterate**

- • Set up dev/test set and metric
- Build initial system quickly
- Use Bias/Variance analysis & Error analysis to prioritize next steps.



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Mismatched training
and dev/test data

Training and testing
on different
distributions

Cat app example

Data from webpages



care about this
Data from mobile app

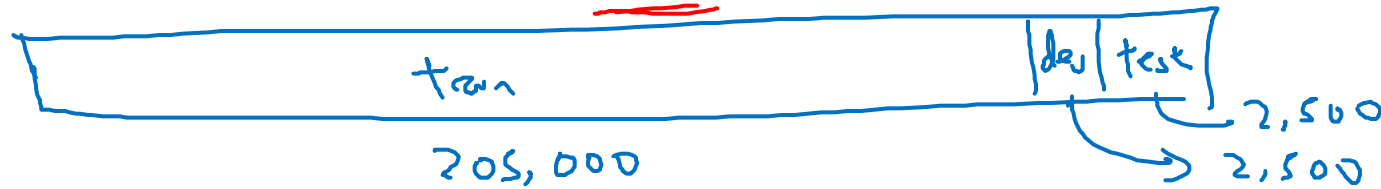


→ ≈ 200,000

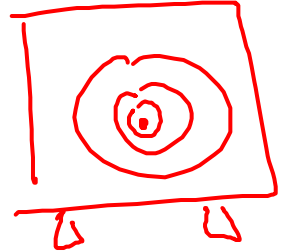
→ 210,000
↓ shuffle

→ ≈ 10,000

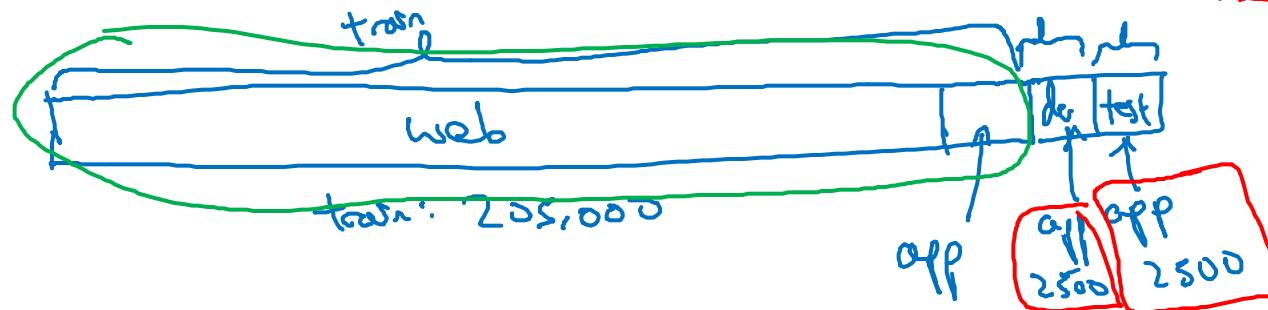
~~Option 1:~~



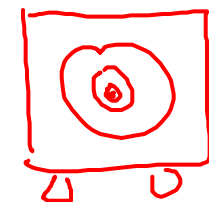
$\frac{200K}{210K}$



Option 2:



2381 - web
119 - mobile app



Speech recognition example

Speech activated rearview mirror



Training

Purchased data

$\downarrow \downarrow$
 X, y

Smart speaker control

Voice keyboard

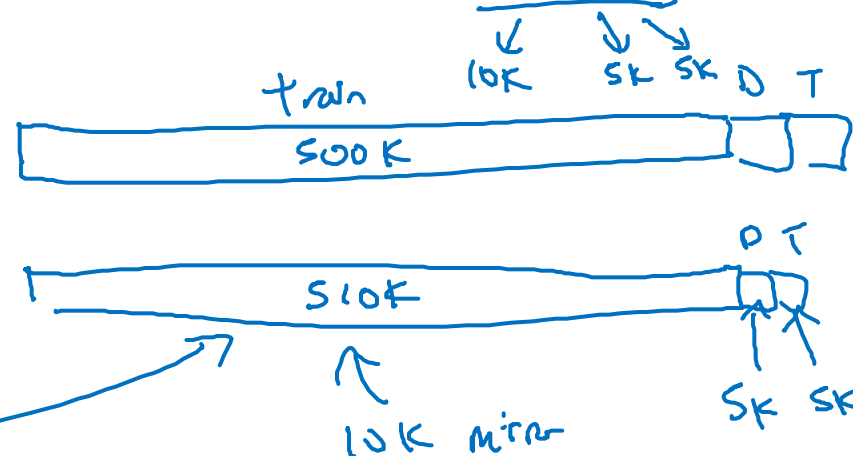
...

500,000 utterances

Dev/test

Speech activated
rearview mirror

→ 20,000





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Mismatched training
and dev/test data

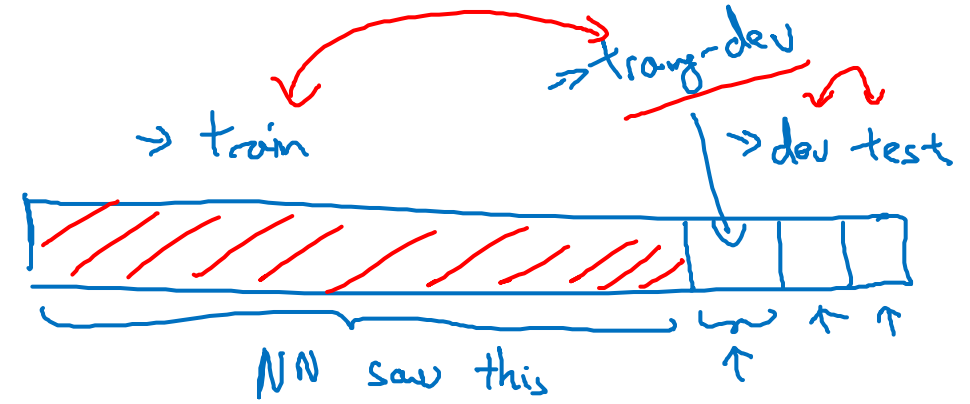
Bias and Variance with
mismatched data
distributions

Cat classifier example

Assume humans get $\approx 0\%$ error.

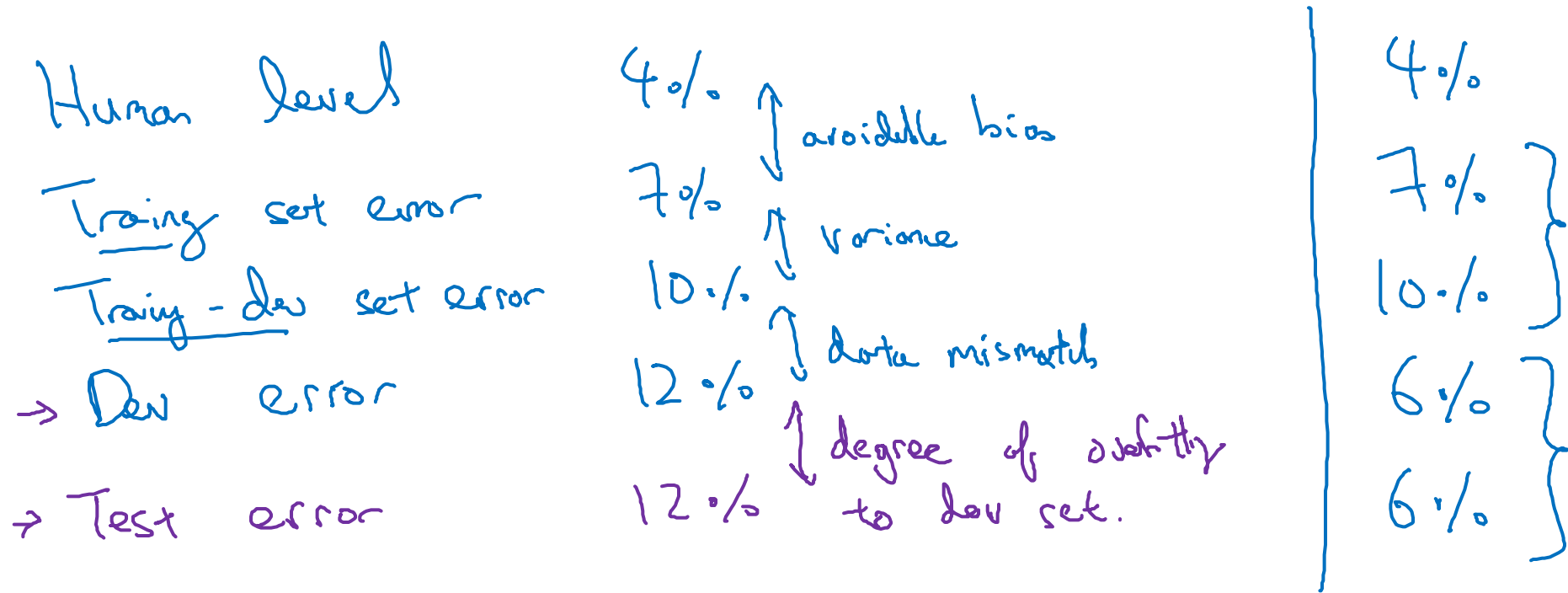
Training error 1%
Dev error 10% $\downarrow 9\%$

Training-dev set: Same distribution as training set, but not used for training



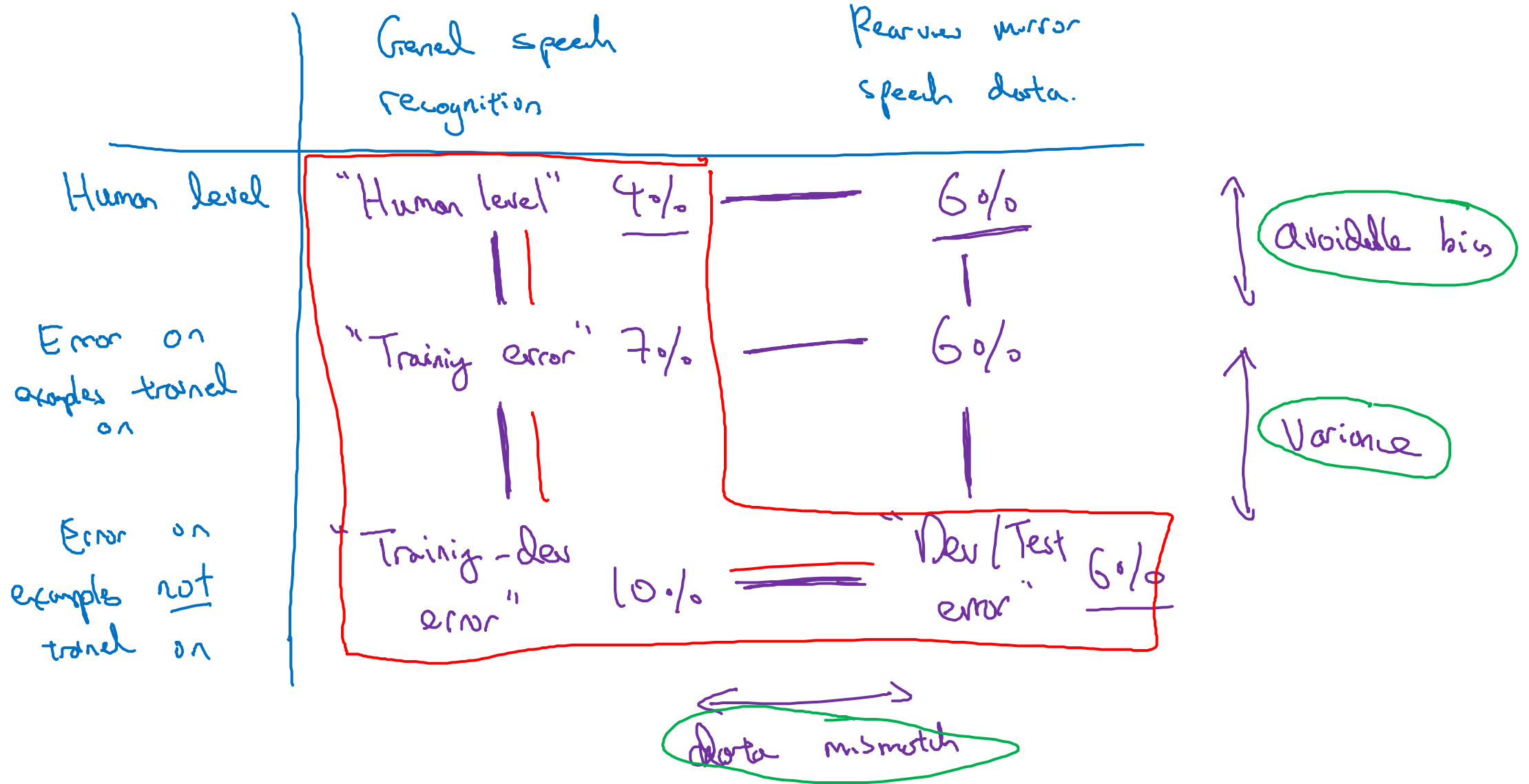
Training error	1%		1%	
→ Training-dev error	9%	↑ variance	1.5%	↑ data mismatch
→ Dev error	10%		10%	
		Variance		
Human error	0%	↑ Avoidable bias	0%	↑ Avoidable bias
Training error	10%	↓ bias	10%	↓ variance
Training-dev error	11%		11%	↑ data mismatch
Dev error	12%		20%	
	Bias		Bias + data mismatch	

Bias/variance on mismatched training and dev/test sets



More general formulation

Recurrent mirror





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Mismatched training
and dev/test data

Addressing data
mismatch

Addressing data mismatch

- • Carry out manual error analysis to try to understand difference between training and dev/test sets

E.g. noisy - car noise

street numbers

- • Make training data more similar; or collect more data similar to dev/test sets

E.g. Simulate noisy in-car data

Artificial data synthesis



+



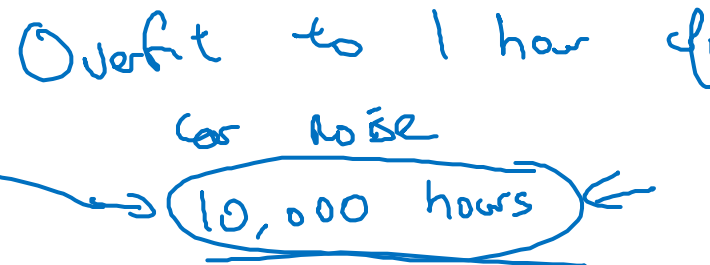
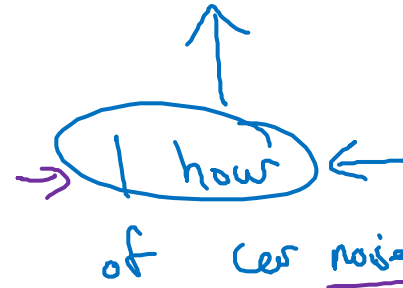
=



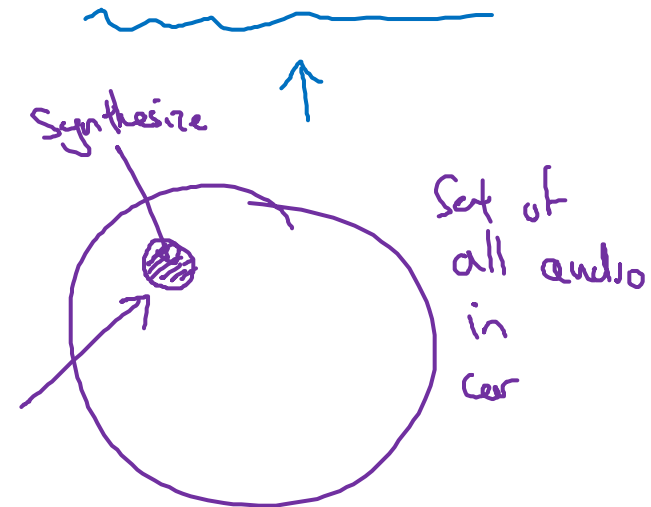
“The quick brown
fox jumps
over the lazy dog.”

↑
10,000 hours

Car noise



Synthesized
in-car audio

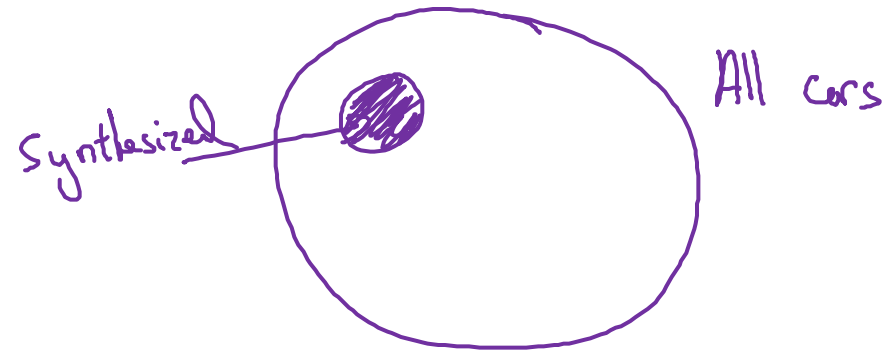


Artificial data synthesis

Car recognition:



≈ 20 cars



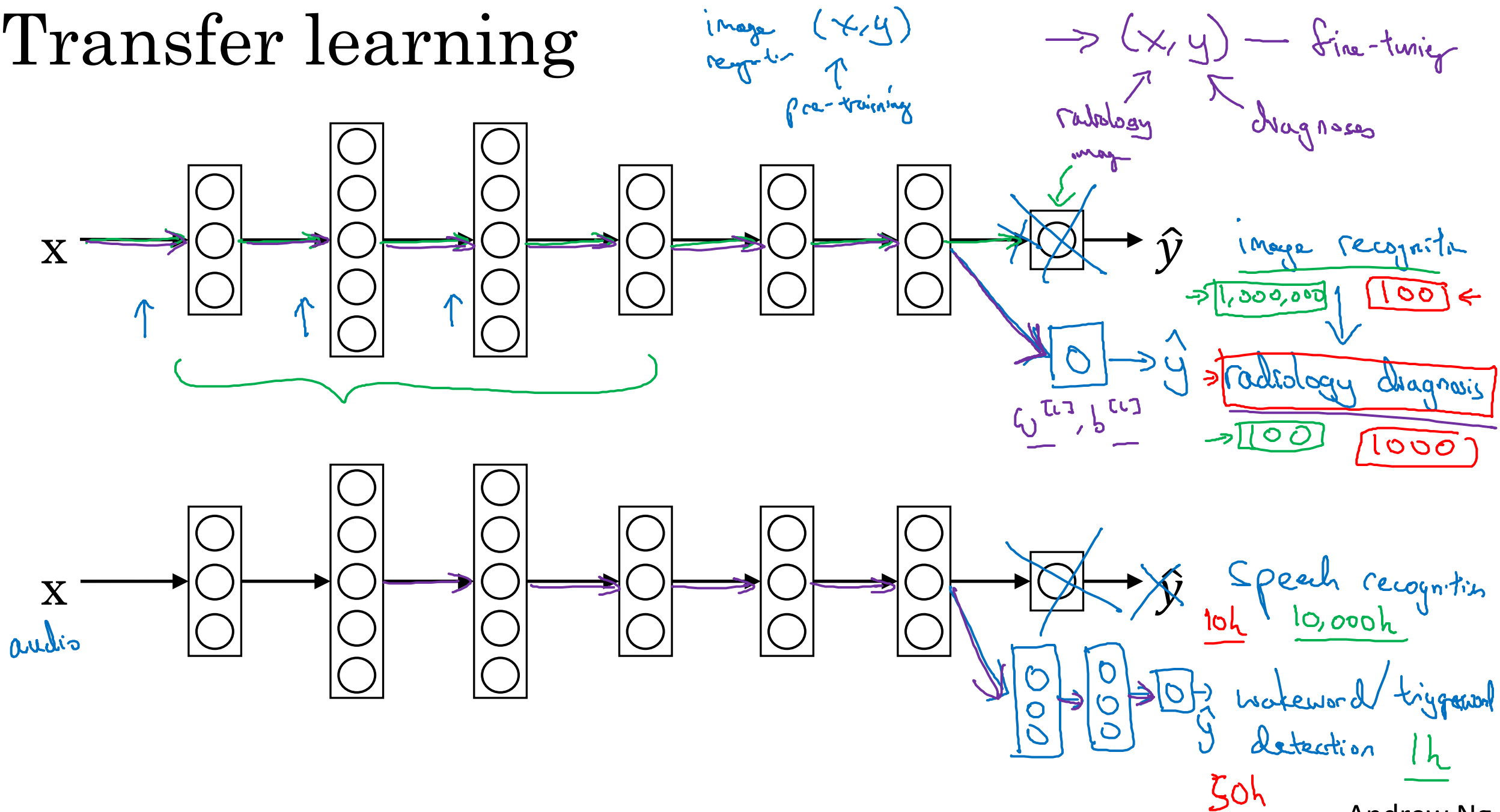


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Learning from
multiple tasks

Transfer learning

Transfer learning



When transfer learning makes sense

Transfer from A $\rightarrow$ B

- Task A and B have the same input x .
- You have a lot more data for Task A than Task B.

- Low level features from A could be helpful for learning B.



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Learning from
multiple tasks

Multi-task
learning

Simplified autonomous driving example



$x^{(i)}$

Pedestrians

Cars

Stop signs

Traffic lights

⋮

$y^{(i)}$

0

1

1

0

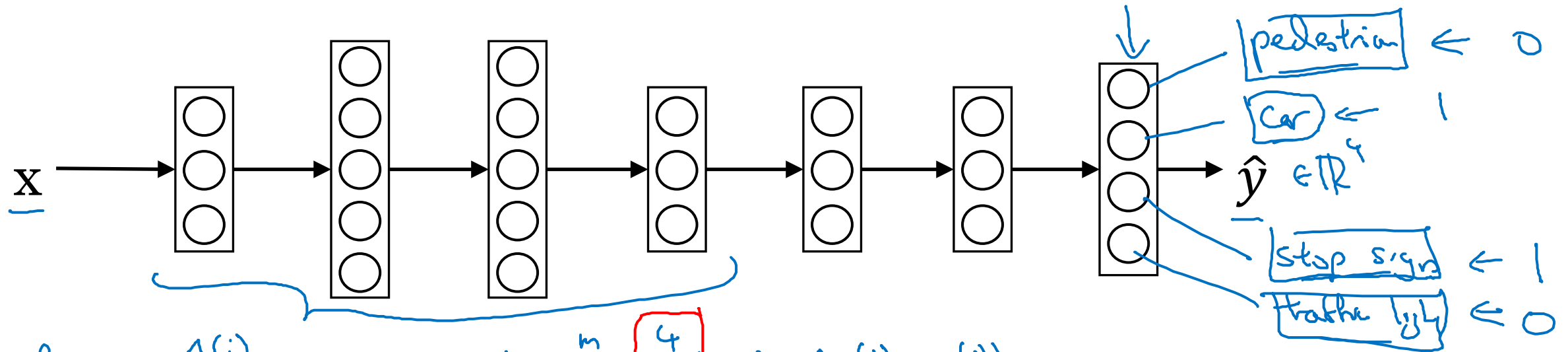
⋮

$(4, 1)$

$$Y = \begin{bmatrix} y^{(1)} & y^{(2)} & y^{(3)} & \dots & y^{(m)} \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix}$$

$(4, m)$

Neural network architecture



Loss: $\frac{1}{m} \sum_{i=1}^m \sum_{j=1}^4 \mathcal{L}(\hat{y}_j^{(i)}, y_j^{(i)})$

Sum only over
value of j with
0/1 label.

Unlike softmax regression:
One image can have multiple labels

Usual logistic loss

$$-y_j^{(i)} \log \hat{y}_j^{(i)} - (1 - y_j^{(i)}) \log (1 - \hat{y}_j^{(i)})$$

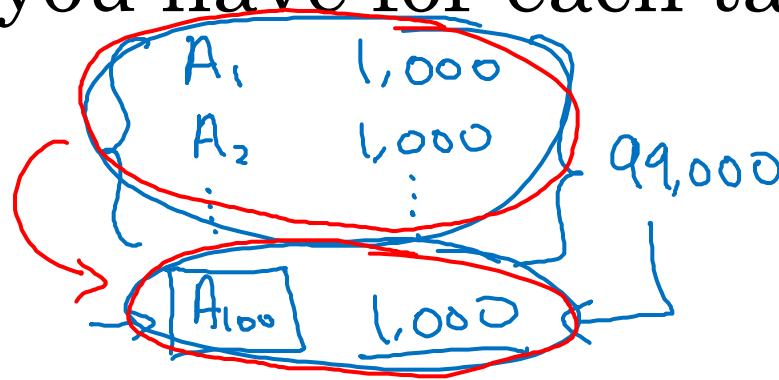
Multi-task learning $\leftarrow$

$$Y = \begin{bmatrix} 1 & 1 & \dots & 1 & ? \\ 0 & 1 & \dots & 1 & ? \\ ? & ? & \dots & 1 & ? \\ ? & ? & \dots & 0 & ? \end{bmatrix}$$

When multi-task learning makes sense

- Training on a set of tasks that could benefit from having shared lower-level features.
- Usually: Amount of data you have for each task is quite similar.

A 1,000,000
↓ ↓
B 1,000



- Can train a big enough neural network to do well on all the tasks.



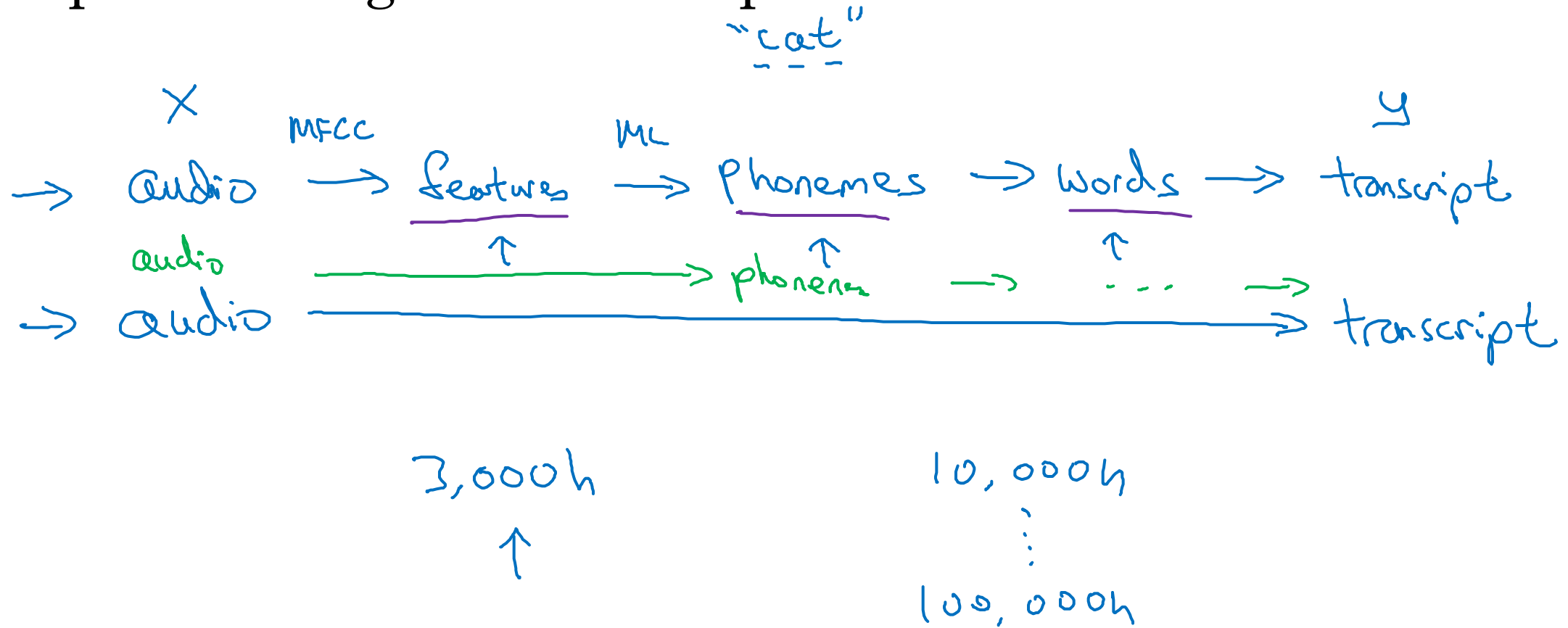
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End-to-end deep learning

What is end-to-end deep learning

What is end-to-end learning?

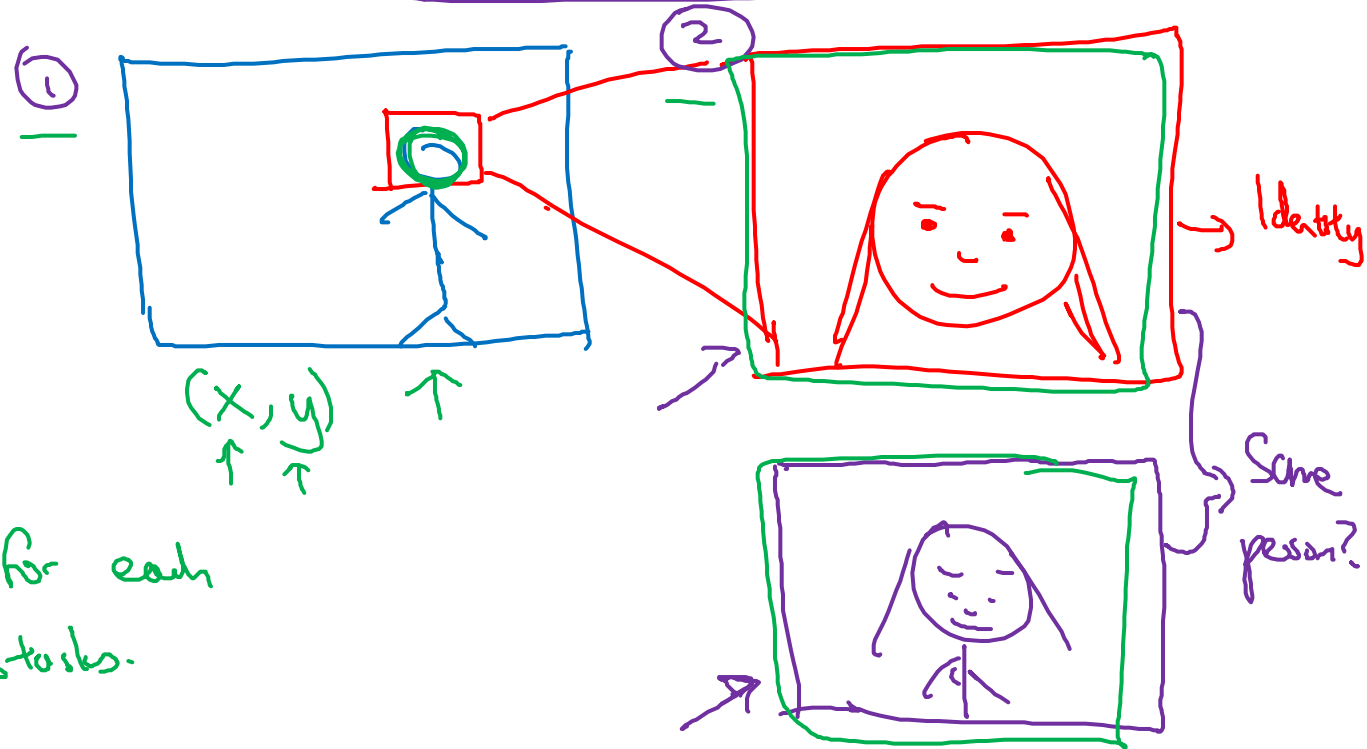
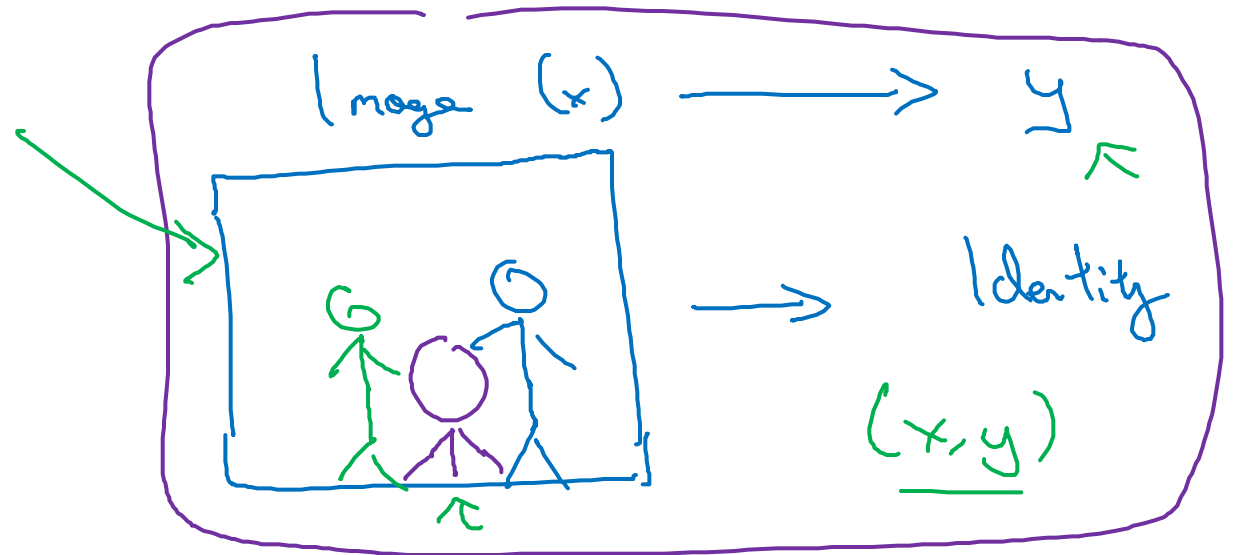
Speech recognition example



Face recognition



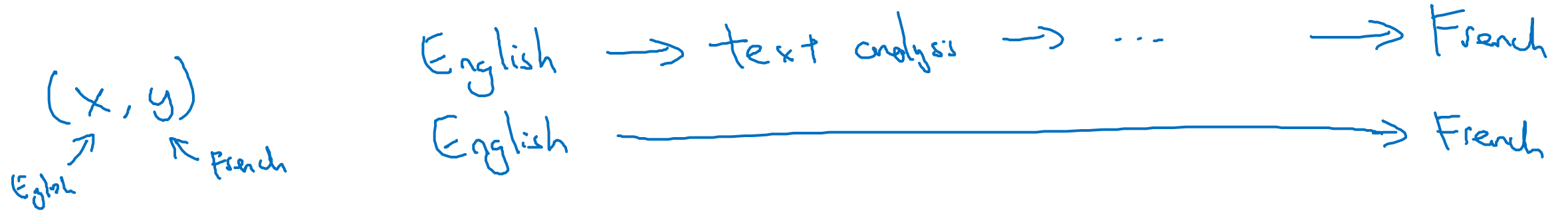
[Image courtesy of Baidu]



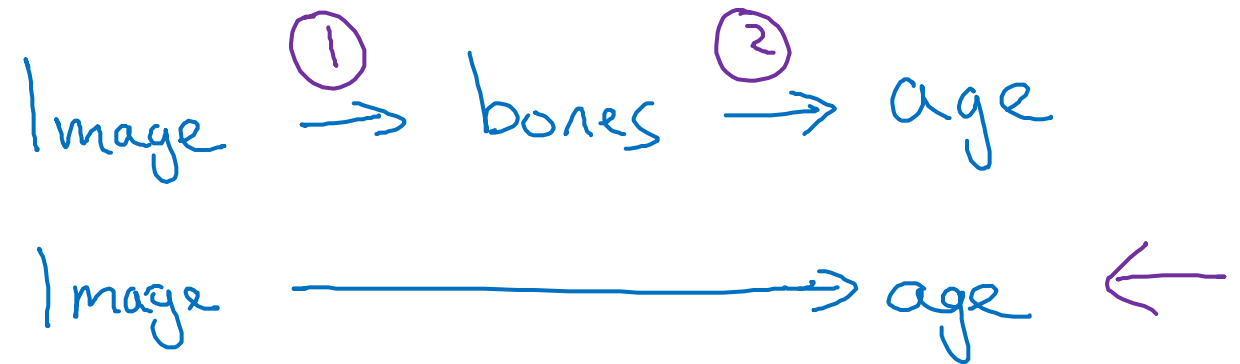
Have data for each
of 2 sub-tasks.

More examples

Machine translation



Estimating child's age:





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End-to-end deep
learning

Whether to use
end-to-end learning

Pros and cons of end-to-end deep learning

Pros:

- Let the data speak
- Less hand-designing of components needed

$x \rightarrow y$

→ "phonemes"
c a t

Cons:

- May need large amount of data
- Excludes potentially useful hand-designed components

$x - - - - - \rightarrow y$

input
end
↓
 $x \rightarrow y$
output
end
↓

(x, y)

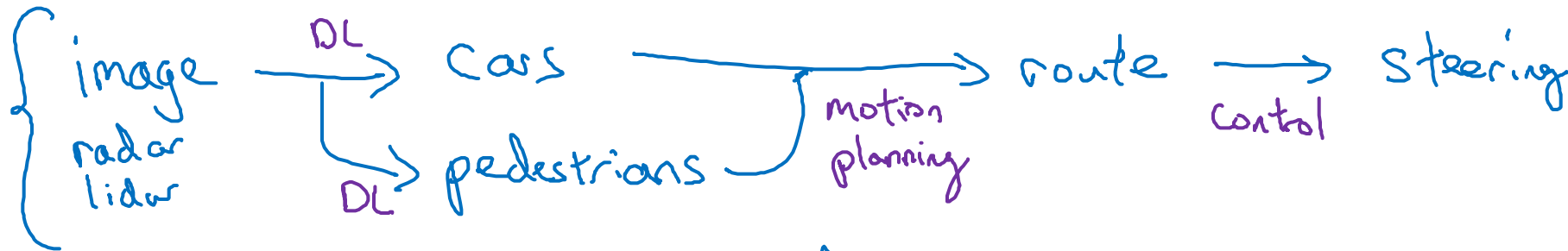
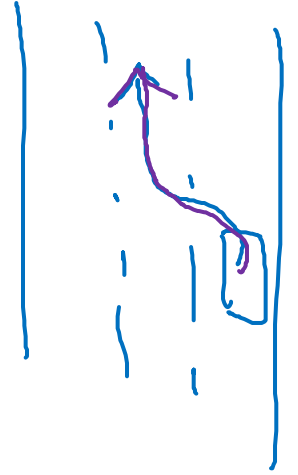
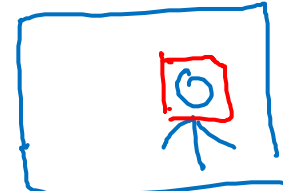
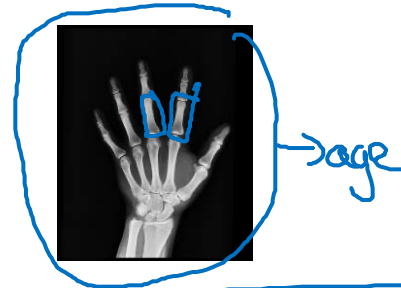
Data.
- - - -

Hand-design.
- - - -

Applying end-to-end deep learning

Key question: Do you have sufficient data to learn a function of the complexity needed to map x to y ?

$x \rightarrow y$



- Use DL to learn individual components
- Carefully choose $x \rightarrow y$ depending what tasks you can get data for.

$\rightarrow$ image $\rightarrow$ steering